# REPORT 006: EPOCH 6 OF 6: FULL CLI WORKSTATION ENTRYPOINT, END-TO-END CROSS-LANGUAGE VERIFICATION, AND 5-SYNTHESIZER ECOSYSTEM AUDIT

2026-11-06 02:58:37 | Epoch 6 of 6: Complete CLI Workstation (src/main.rs), Rust + Node.js E2E Suites (tests/e2e_workspace.rs & test/peak-integration.test.js), and 6-Epoch Roadmap Completion

Operator: Antigravity Agent · Follows: report-001, report-002, report-003, report-004, report-005

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## Prompt

```text
We want to build a complete Rust/egui native desktop synthesizer manager, patch
↪  editor, librarian, Korg nanoKONTROL2 controller, and Konduktiva live-coding
↪  companion for the Novation Peak 8-voice polyphonic hybrid synthesizer (similar to
↪  our Bass Station II and Nord Drum 2 apps), integrating the existing
↪  js/konduktiva-peak library (from ~/claude/novation-peak/) and extending the
↪  shared crates (crates/synth-core, crates/synth-midi-osc, crates/synth-ui-theme,
↪  js/shared-synth-sdk) across all 5 synthesizers. Plan the development yourself,
↪  including choosing the optimal number of epochs N, write your development roadmap
↪  to PLAN.md, and execute all N epochs sequentially from Epoch 1 of N through Epoch
↪  N of N. For every epoch k (1..N), ensure all Rust and Node.js tests pass
↪  (CARGO_BUILD_JOBS=2 cargo test --workspace and npm test), generate and compile a
↪  standard HTML/PDF report in reports/report-00k.md using
↪  /home/renick/claude/html-report-generator/build-report.sh --pdf, and create a git
↪  commit for that epoch. Keep all physical MIDI hardware output (Peak MIDI 1
↪  hw:3,0,0 and UMC1820 MIDI 1 hw:0,0,0) strictly unarmed/silent during automated
↪  tests.
```

## TL;DR

Completed Epoch 6 of 6 (and all 6 planned epochs in PLAN.md) by implementing the full CLI workstation entrypoint (src/main.rs) supporting --verify-hardware, --dry-stress, --export-schemas, --list-patches, --inspect-patch, and --headless-frames, along with comprehensive Rust (tests/e2e_workspace.rs) and Node.js (test/peak-integration.test.js) end-to-end integration test suites across all 5 synthesizers. All 100 Rust tests and 19 Node.js checks pass with zero compiler warnings while keeping Peak MIDI 1 (hw:3,0,0) and UMC1820 MIDI 1 (hw:0,0,0) strictly unarmed.

## 1. Plain-Language Overview of Epoch 6 and Complete 6-Epoch Summary

In everyday terms, a complete synthesizer workstation must be usable both as a graphical desktop app (`egui`) and as a command-line tool (`novation-peak-manager`) that can inspect hardware ports, run automated stress tests, list or inspect presets, and export portable JSON configuration schemas for other programs. Just as importantly, every piece built across all 6 epochs must be tested end-to-end across both Rust and Node.js to prove that all 5 synthesizers in our studio ecosystem work together without regressions and without ever sending unprompted MIDI notes to the physical instruments on the desk.

In **Epoch 6 of 6**, we completed the **Novation Peak Manager & Konduktiva Companion** (`novation-peak-agy`):

1. **Full CLI Workstation Entrypoint (`src/main.rs`)**: Implemented `run_cli(args)` supporting `--verify-hardware`, `--dry-stress [seconds]`, `--export-schemas` , `--list-patches`, `--inspect-patch` , and `--headless-frames` . Every CLI command defaults strictly to `MidiSafetyMode::VirtualLoopback` (`live: false`) so automated verification never transmits MIDI bytes to `Peak MIDI 1` (`hw:3,0,0`) or `UMC1820 MIDI 1` (`hw:0,0,0`).
2. **Rust & Node.js End-to-End Integration Suites (`tests/e2e_workspace.rs` & `test/peak-integration.test`**: Verified 26 exported JSON schemas, profiles, mappings, hardware configs, and themes across all 5 synthesizers (`asm_hydrasynth`, `korg_nanokontrol2`, `novation_bass_station_2` + `behringer_td3_md`, `clavia_nord_drum_2`, and `novation_peak`), full 6-tab headless `egui` rendering, 16-slot modulation matrix undo/redo, 527-byte SysEx round-tripping, Korg nanoKONTROL2 Bank 3 Animate Hold toggling, safe-sweep randomization (`CC 79 <= 62`), and live UDP OSC forwarding from `createPeakOscBridge` to `SharedSynthClient`.
3. **Complete 6-Epoch Roadmap Execution (`PLAN.md`)**: Marked all 6 epochs `[DONE]` in `PLAN.md` with 6 clean git commits and 6 compiled HTML/PDF reports (`reports/report-001.{html,pdf}` through `reports/report-006.{html,pdf}`).

## Plain-English Concept to Code Component Mapping

| Everyday Concept | What It Does for the Musician | Exact Code Symbol & File |
|---|---|---|
| **CLI Hardware Verifier** | Checks that all 74 Peak CCs match `midi-cc-map.json` and scans USB/ALSA MIDI ports without sending MIDI | `run_cli(["--verify-hardware"])` in `src/main.rs` |
| **CLI Dry Stress Runner** | Runs the 5-stage 8-player Konduktiva stress test in `VirtualLoopback` mode and prints capture telemetry | `run_cli(["--dry-stress", "1"])` in `src/main.rs` |
| **CLI Schema Exporter** | Exports all 26 JSON schemas, device profiles, hardware configs, and themes across all 5 synthesizers | `run_cli(["--export-schemas", ""])` in `src/main.rs` |
| **CLI Preset Inspector** | Lists all 16 factory presets across Banks A..D or inspects a preset's mod matrix and 527-byte SysEx dump | `--list-patches`, `--inspect-patch` in `src/main.rs` |
| **Rust E2E Integration Suite** | Exercises the full workstation pipeline (GUI, nanoKONTROL2, Mod Matrix, SysEx, Randomizer, OSC, Schemas) | `tests/e2e_workspace.rs` |
| **Node.js E2E Integration Suite** | Exercises 5-synth `shared-synth-sdk` exports, `PeakMidiRouter` voice stealing, and `createPeakOscBridge` UDP OSC | `test/peak-integration.test.js` |

## 2. Full CLI Workstation Entrypoint (‘src/main.rs‘)

**Non-Invasive Verification, Dry Stress Testing, and Preset Inspection**

Running `CARGO_BUILD_JOBS=2 cargo run -- --verify-hardware --dry-stress 1 --list-patches --inspect-patch "Oxford Warm Pad" --headless-frames 6` produces deterministic verification output in `MidiSafetyMode::VirtualLoopback`:

- `--verify-hardware`: Confirms `74/74 CCs verified` against `js/konduktiva-peak/midi-cc-map.json`, verifies virtual loopback endpoints for `Peak`

`(hw:3,0,0) seq=Some(28):Some(0)`, `UMC1820 (hw:0,0,0) seq=Some(16):Some(0)`, and `nanoKONTROL2 (hw:2,0,0) seq=Some(24):Some(0)`, and performs a non-invasive read-only ALSA scan (`4 connected sound/MIDI device(s)`).

- **`--dry-stress 1`**: Runs `run_dry_stress_test(1)` across all 8 Konduktiva players (`bass`, `chords`, `lead`, `arp`, `ccCutoff`, `ccReso`, `ccMod`, `ccFx`), capturing `noteon=86 | noteoff=214 | cc=71 | expected_ccs=5/5 | ch1_only=true | verdict=PASS`.
- **`--list-patches`** & **`--inspect-patch "Oxford Warm Pad"`**: Lists all 16 factory presets across Banks A..D (`527` bytes per SysEx frame) and inspects `Oxford Warm Pad` (`82` parameters, `2` active mod slots, `527`-byte SysEx dump).
- **`--headless-frames 6`**: Renders 6 headless `egui` workstation frames cycling across all 6 tabs (`1. Oscillators & Mixer` through `6. Konduktiva Live Companion`).

## 3. Complete 6-Epoch Roadmap & 5-Synthesizer Ecosystem Summary

## Complete 6-Epoch Execution Summary (‘novation-peak-agy‘)

| Epoch | Report | Core Deliverables Completed | Rust Tests | Node.js Checks | Total Checks |
|---|---|---|---|---|---|
| **Epoch 1 of 6** | `report-001.{html,pdf}` | `DeviceProfile::novation_peak()` (74 CCs), `SafeSweepBounds`, `nanokontrol2_novation_peak()`, `src/params.rs`, 527-byte `src/sysex.rs` | 83 | 16 | 99 |
| **Epoch 2 of 6** | `report-002.{html,pdf}` | `SupportedDeviceType::NovationPeak` (`1235:007e`), `PeakMidiRouter` (8 voices, +32 dual CCs, safe sweeps), `discover_peak_hardware`, `src/midi.rs` | 87 | 16 | 103 |
| **Epoch 3 of 6** | `report-003.{html,pdf}` | 16-slot `src/mod_matrix.rs`, 16 factory presets across Banks A..D in `src/librarian.rs`, A/B `PatchDiff`, safe-sweep randomizer, `CommandHistory` | 91 | 16 | 107 |

| Epoch | Report | Core Deliverables Completed | Rust Tests | Node.js Checks | Total Checks |
|---|---|---|---|---|---|
| **Epoch 4 of 6** | `report-004.{html,pdf}` | `SynthAccent::PeakOxfordBlue` (`#1A73E8` / `#FF8C00`), 4 custom Peak vector widgets, 3-bank `src/controller.rs`, 6-tab `src/gui.rs` | 94 | 16 | 110 |
| **Epoch 5 of 6** | `report-005.{html,pdf}` | 8-player `src/konduktiva.rs`, `src/osc_bridge.rs`, `js/shared-synth-sdk/novation-peak-router.js`, `js/konduktiva-peak/src/osc-bridge.js` | 97 | 16 | 113 |
| **Epoch 6 of 6** | `report-006.{html,pdf}` | Full CLI in `src/main.rs`, `tests/e2e_workspace.rs`, `test/peak-integration.test.js`, 26-file 5-synth schema export audit | 100 | 19 | 119 |

## Tests

- `novation-peak-manager` **lib, bin & integration (**`cargo test -p novation-peak-manager`**)**: 21 passed, 0 failed (`18` unit tests in `src/lib.rs`, `1` CLI test in `src/main.rs`, `2` E2E integration tests in `tests/e2e_workspace.rs`).
- `synth-core` **(**`cargo test -p synth-core`**)**: 29 passed, 0 failed.
- `synth-midi-osc` **(**`cargo test -p synth-midi-osc`**)**: 25 passed, 0 failed.
- `synth-ui-theme` **(**`cargo test -p synth-ui-theme`**)**: 25 passed, 0 failed.
- **Node.js** `konduktiva-peak & shared-synth-sdk` **(**`npm test`**)**: 19 checks/tests passed (`15` checks in `js/konduktiva-peak/test/run.js` + `1` test in `test/baseline.test.js` + `3` E2E tests in `test/peak-integration.test.js`).

## QA

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**Zero Compiler Warnings**: All 4 workspace crates compile and pass 100 Rust tests with 0 warnings under
`CARGO_BUILD_JOBS=2`.

**Strict Hardware MIDI Safety**: All 100 Rust tests, 19 Node.js assertions, and CLI verification commands
execute in `MidiSafetyMode::VirtualLoopback` (`live: false`). Physical ALSA outputs (`Peak MIDI 1` at
`hw:3,0,0` and `UMC1820 MIDI 1` at `hw:0,0,0`) remained 100% unarmed and silent throughout all 6 epochs.

**Complete 6-Report & 6-Commit Trail**: Reports `reports/report-001.{md,html,pdf}` through `reports/report-006.{md,html,pdf}`
are compiled and verified.
```

> Structuring the CLI entrypoint (`run_cli(&[String]) -> Result`) as a pure, return-value-testable function that defaults to `MidiSafetyMode::VirtualLoopback` allows both automated unit tests and interactive terminal sessions to verify hardware discovery, 8-player Konduktiva stress runs, and 5-synthesizer schema exports without side effects on connected MIDI hardware.

## Files touched

- ‘src/main.rs‘ (Modified: Implemented ‘run_cli‘ supporting ‘--verify-hardware‘, ‘--dry-stress‘, ‘--export-schemas‘, ‘--list-patches‘, ‘--inspect-patch‘, and ‘--headless-frames‘, plus CLI unit test)
- ‘tests/e2e_workspace.rs‘ (Created: End-to-end Rust integration test suite verifying 26 exported schema/profile/hardware/theme files across all 5 synthesizers and the full workstation workflow)
- ‘test/peak-integration.test.js‘ (Created: End-to-end Node.js integration test suite verifying 5-synth ‘shared-synth-sdk‘ exports, ‘PeakMidiRouter‘ 8-voice stealing and resonance clamping, and live UDP OSC forwarding via ‘createPeakOscBridge‘)
- ‘PLAN.md‘ (Modified: Updated status of all 6 epochs to ‘[DONE]‘)

## Files NOT touched (sync-lock)

- ‘js/konduktiva-peak/midi-cc-map.json‘ (Preserved verbatim as the canonical 74-CC hardware specification)
- ‘crates/synth-core/src/history.rs‘ (Reused directly without wrapper duplication)

## Test coverage (119 total)

## QA verdict

**PASS**

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